Generic local identifiability does not imply generic global identifiability even when $\rho(G) < 1$

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Abstract

We give a twelve-node network with zero diagonal and forty-two independently unknown edge entries. Nine nodes are excited and nine are measured by fixed selection matrices. The observation map $G \mapsto C(I-G)^{-1}B$ is generically locally identifiable. A nonempty open subset of the admissible parameter space has two distinct compatible networks, both satisfying $\rho(G) < 1$. The counterexample therefore disproves generic local implying generic global identifiability even with this spectral-radius restriction. An explicit rational pair and exact matrix-power bounds certify the claim. The model uses no prescribed equalities between unknown edges, no known nonzero edges, and no self-loops.

Scope and the meaning of generic. We consider $w = Gw+Br$, $y = Cw$, with known binary node selections B, C, zero diagonal of G, and independently unknown entries at its permitted off-diagonal positions. Impose $\rho(G) < 1$ on every admissible candidate, including any alternative network. The stronger assumption appears in the later Legat–Hendrickx manuscript [2]. The matrices in the explicit pair are real constants, so they also give proper, stable rational transfer functions satisfying the spectral-radius restriction at every frequency. In the supplied paper's notation, $P = I - G$, $Q = B$, $R = C$, $S = 0$.

Generic local identifiability means that the observation Jacobian has full column rank outside a proper algebraic subset. Generic global identifiability on the restricted domain would require uniqueness for almost every admissible parameter. A nonempty Euclidean open set of ambiguous admissible parameters disproves this property. We do not claim that every stable parameter in the structure has two stable alternatives.

The six-node ambiguity used in the construction. Let

$$A = \begin{pmatrix} 0 & 0 & a_3 \\ a_1 & 0 & 0 \\ 0 & a_2 & 0 \end{pmatrix}, \quad D = \begin{pmatrix} 0 & d_1 & 0 \\ 0 & 0 & d_2 \\ d_3 & 0 & 0 \end{pmatrix}, \quad X = \operatorname{diag}(x_1, x_2, x_3),$$

and

$$G_b = \begin{pmatrix} A & 0 \\ X & D \end{pmatrix}, \qquad B_b = \begin{pmatrix} I_3 \\ 0 \end{pmatrix}, \qquad C_b = \begin{pmatrix} 0 & I_3 \end{pmatrix}.$$

The nine quantities a_i, d_i, x_i are independent. Its observation is

$$F = C_b(I_6 - G_b)^{-1}B_b = (I_3 - D)^{-1}X(I_3 - A)^{-1}.$$

Set $u_i = a_i/x_i$, $v_i = d_i/x_i$, $z_i = 1/x_i$. The six off-diagonal entries of F^{-1} determine u_i, v_i directly, and its diagonal entries are

$$e_i = z_i + \frac{u_{i-1}v_{i-1}}{z_{i-1}}, \qquad i \pmod 3. \tag{1}$$

With u, v fixed, the determinant of the Jacobian of these three equations with respect to z is

$$\det J_z = 1 - \prod_{i=1}^{3} a_i d_i. \tag{2}$$

It is nonzero generically. The three equations in (1) compose to a Möbius fixed-point equation, hence a quadratic for z_1.

For completeness, a rational formula for the alternative is

$$k_i = a_i d_i \frac{x_{i+1}}{x_i}, \quad h_i = 1 + k_i + k_i k_{i+1}, \quad \eta_i = \frac{k_i h_{i+1}}{h_i}, \quad (a_i', d_i', x_i') = \frac{(a_i, d_i, x_i)}{\eta_i}. \tag{3}$$

The transformation leaves u_i, v_i fixed and sends z_i to $\eta_i z_i$. Writing $L = \prod_i k_i = \prod_i a_i d_i$, the identities

$$\eta_i - 1 = \frac{L-1}{h_i}, \qquad \eta_i + \frac{k_{i-1}}{\eta_{i-1}} = 1 + k_{i-1}$$

show that it preserves all observations and gives a distinct second point when $L \neq 1$. It is well-defined and continuous near the points used below.

Augmentation that preserves all fibers. Add six nodes with signal h, all of which are excited and measured. Allow feedback through matrices $K \in \mathbb{R}^{3\times 6}$, $L_c \in \mathbb{R}^{6\times 3}$ and $J \in \mathbb{R}^{6\times 6}$:

$$\mathcal{G} = \begin{pmatrix} G_b & B_b K \\ L_c C_b & J \end{pmatrix}, \qquad \mathcal{B} = \begin{pmatrix} B_b & 0 \\ 0 & I_6 \end{pmatrix}, \qquad \mathcal{C} = \begin{pmatrix} C_b & 0 \\ 0 & I_6 \end{pmatrix}. \tag{4}$$

Thus $\mathcal{B}$ excites nodes $1, 2, 3, 7, 8, 9, 10, 11, 12$, and $\mathcal{C}$ measures nodes $4, 5, 6, 7, 8, 9, 10, 11, 12$.

Lemma 1. *Where the displayed inverses exist, the augmented observation $H = \mathcal{C}(I_{12} - \mathcal{G})^{-1}\mathcal{B}$ satisfies*

$$H^{-1} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix}. \tag{5}$$

Consequently, the observation determines every permitted entry of K, L_c, J, and its remaining ambiguity is precisely the ambiguity in G_b.

Proof. Let the original and new excitations be r and s. The node equations imply $y = F(r + Kh)$ and $h = L_c y + Jh + s$. Rearranging gives

$$\begin{pmatrix} r \\ s \end{pmatrix} = \begin{pmatrix} F^{-1} & -K \\ -L_c & I_6 - J \end{pmatrix} \begin{pmatrix} y \\ h \end{pmatrix},$$

which proves the identity and its identifiability consequences. □

An explicit pair with small spectral radii. Let

$$U = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \qquad V = U + U^T, \qquad I = I_3.$$

Choose the numerical values

$$K_0 = \begin{pmatrix} V & -U \end{pmatrix}, \qquad L_0 = \begin{pmatrix} U^T \\ -V \end{pmatrix}, \qquad J_0 = \begin{pmatrix} U^T & -V \\ I & U \end{pmatrix}.$$

For scalars r, x, define the following matrix in 3×3 blocks:

$$M(r,x) = \begin{pmatrix} -rU & 0 & V & -U \\ xI & -rU^T & 0 & 0 \\ 0 & U^T & U^T & -V \\ 0 & -V & I & U \end{pmatrix}. \tag{6}$$

Now take

$$r = \frac{99}{100}, \qquad \mathcal{G}_1 = M(r,1), \qquad \mathcal{G}_2 = M(r^{-1}, r^{-2}). \tag{7}$$

Both matrices have exactly the same zero pattern and zero diagonal. The model assigns a separate independent unknown to *each* of the forty-two permitted entries of this pattern: nine in G_b, nine in K, nine in L_c and fifteen in J. The block notation in (6) gives convenient values for two points of this full model; it imposes no equality constraints on the model's unknowns and fixes no nonzero edge in advance.

Theorem 1. *The forty-two-parameter network structure just defined is generically locally identifiable but is not generically globally identifiable on the domain $\rho(\mathcal{G}) < 1$.*

Proof. Equality of observations. For the original six-node block in $M(r,x)$,

$$F(r,x)^{-1} = x^{-1}(I + rU)(I + rU^T) = x^{-1}\big[(1+r^2)I + rV\big].$$

Therefore $F(r,1) = F(r^{-1}, r^{-2})$. Since both augmented networks use the same K_0, L_0, J_0, Lemma 1 proves that their complete 9×9 observations are equal. In fact their common inverse observation is

$$H^{-1} = \begin{pmatrix} (1+r^2)I + rV & -V & U \\ -U^T & I - U^T & V \\ V & -I & I - U \end{pmatrix}.$$

Exact spectral-radius certificates. Direct rational multiplication of the explicit matrices gives

$$\|\mathcal{G}_1^4\|_\infty = \frac{33910001}{100000000} < 1, \qquad \|\mathcal{G}_2^4\|_\infty = \frac{16976126}{32019867} < 1. \tag{8}$$

Here $\|\cdot\|_\infty$ is the maximum absolute row sum. Hence

$$\rho(\mathcal{G}_i)^4 = \rho(\mathcal{G}_i^4) \le \|\mathcal{G}_i^4\|_\infty < 1,$$

proving the required restriction for both candidates without relying on floating-point eigenvalues. Their numerical spectral radii are approximately 0.509563 and 0.404210, respectively.

Generic local identifiability. In (5), the thirty-three controller entries occur separately and linearly. The remaining 3×3 block F^{-1} has generic Jacobian rank nine by (2). Consequently the full observation Jacobian has generic rank $9 + 33 = 42$. It has rank forty-two at both explicit points, since $\prod_i a_i d_i = r^6$ or r^{-6}, neither of which is one.

The ambiguity occurs on an open admissible set. Apply (3) to the nine independent entries of G_b, and keep the thirty-three independent entries of K, L_c, J unchanged. At the first point in (7), this maps to the second point. The formulas are rational and continuous on a neighborhood of that point. The two networks are distinct and both inequalities (8) are strict, so a sufficiently small open neighborhood in the full forty-two-dimensional parameter space consists of distinct pairs satisfying $\rho < 1$ and having identical observations. This set has positive measure and cannot be contained in a proper algebraic subset. Thus generic global identifiability fails even after restricting both candidates to $\rho < 1$. □

Why the augmentation can make both branches stable. The two six-node branches coalesce at $r = 1$. Their common augmented matrix satisfies

$$M(1,1)^4 = 0. \tag{9}$$

This can be understood without multiplying twelve-dimensional matrices. At $r = x = 1$, let G_{b0} denote the six-node block. The controller satisfies $J_0 = G_{b0} + B_b K_0 - L_0 C_b$. A change of coordinates from (w, h) to $(w, w-h)$ makes the augmented matrix block upper triangular, with diagonal blocks

$$G_{b0} + B_b K_0 = \begin{pmatrix} U^T & -U \\ I & -U^T \end{pmatrix}, \qquad G_{b0} - L_0 C_b = \begin{pmatrix} -U & -U^T \\ I & U \end{pmatrix}.$$

Each of these squares to zero, proving (9). By continuity, both $M(r,1)$ and $M(r^{-1}, r^{-2})$ have arbitrarily small spectral radii when $r \neq 1$ is sufficiently close to one. The ambiguity can therefore occur with $\rho < \varepsilon$ for any prescribed $\varepsilon > 0$. At $r = 1$ the local rank drops; the counterexamples use $r \neq 1$.

Reproducible checks. The script `verify_spectral_counterexample.py` checks: the common zero-diagonal pattern and edge count; equality of the complete rational 9×9 observations; the block inverse identity; the exact norms in (8); and nilpotence in (9). It also forms the full 81×42 observation Jacobian and certifies rank forty-two at both points by elimination modulo the prime 1000003. A nonzero minor modulo this prime certifies a nonzero rational minor. All denominators used in this certificate are invertible modulo the prime.

As a check against accidental equality constraints, the script independently perturbs all forty-two edge values to distinct rational numbers, applies the rational alternative formula, and verifies exact observation equality and both strict matrix-power bounds again. The detailed results and explicit matrices are saved in `spectral_verification_results.json`. The proof above is independent of the numerical search used while exploring other possible constructions. No minimality or literature-priority claim is made.

References

[1] A. Legat and J. M. Hendrickx, "Local Network Identifiability with Partial Excitation and Measurement," *59th IEEE Conference on Decision and Control*, 2020, pp. 4342–4347. Author's PDF.

[2] A. Legat and J. M. Hendrickx, "Identifiability in Networked Systems: Algebraic and Combinatorial Conditions," manuscript listed by the author as submitted in August 2024. Author's PDF.